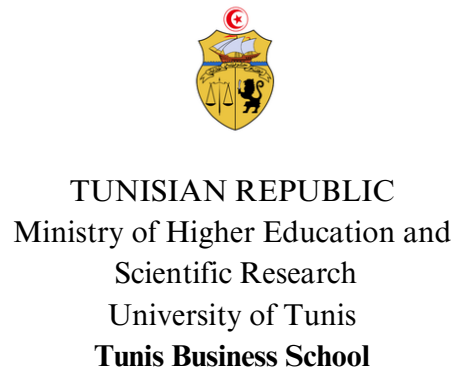




THE GEOPOLITICS OF OIL: ASSESSING THE MACROECONOMIC IMPACT OF THE 2026 GULF CRISIS

A Panel Data Econometrics Analysis Project

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1 Data

1.1 Loading and Preparing the Data

```
# Uncomment to install if needed:
# install.packages(c("tidyverse", "plm", "lmtest", "stargazer",
#                    "ggplot2", "car", "forecast", "tseries", "kableExtra", "scales"))

library(tidyverse)
library(plm)
library(lmtest)
library(stargazer)
library(ggplot2)
library(car)
library(forecast)
library(tseries)
library(zoo)
library(kableExtra)
library(scales)

# Load the dataset - make sure the CSV is in your working directory
data_raw <- read.csv("panel_data_8countries.csv", stringsAsFactors = FALSE)

cat("Dimensions:", nrow(data_raw), "rows x", ncol(data_raw), "columns\n")

## Dimensions: 336 rows x 8 columns

glimpse(data_raw)

## Rows: 336
```

```
## Columns: 8
## $ country      <chr> "Canada", "Canada", "Canada", "Canada", "Canada", "Canada"~
## $ date         <chr> "01/01/2023", "01/02/2023", "01/03/2023", "01/04/2023", "0~
## $ cpi          <dbl> 119.8000, 120.0200, 120.2400, 120.4600, 120.6800, 120.9000~
## $ ipi          <dbl> 106.5000, 106.6000, 106.7000, 106.8000, 106.9000, 107.0000~
## $ brent        <dbl> 83.96, 83.16, 79.49, 84.99, 75.23, 74.89, 81.63, 85.77, 93~
## $ neer         <dbl> 93.5000, 93.3455, 93.1909, 93.0364, 92.8818, 92.7273, 92.5~
## $ policy_rate  <dbl> 4.2500, 4.4000, 4.5500, 4.7000, 4.8500, 5.0000, 5.0000, 5.~
## $ vix          <dbl> 19.98, 20.74, 19.95, 16.77, 16.73, 13.59, 13.72, 15.68, 17~
```

```
# Define country groups
```

```
emerging_importers <- c("Turkey", "Poland", "India",
                        "Indonesia", "South_Africa", "Thailand")
advanced_exporters <- c("United_States", "Canada")
```

```
# Process data: parse dates, compute log variables, post-crisis dummy
```

```
data_processed <- data_raw %>%
  mutate(
    date      = as.Date(date, format = "%d/%m/%Y"),
    ln_cpi    = log(cpi),
    ln_ipi    = log(ipi),
    ln_neer   = log(neer),
    post_crisis = as.integer(date >= as.Date("2026-03-01")),
    oil_balance = if_else(country %in% advanced_exporters,
                          "Exporter", "Importer"),
    dev_level  = if_else(country %in% advanced_exporters,
                          "Advanced", "Emerging"),
    group      = paste(dev_level, oil_balance, sep = "_")
  )
```

```
# Compute oil shock from Brent series (recalculated, not taken from CSV)
```

```

oil_shock_lookup <- data_processed %>%
  distinct(date, brent) %>%
  arrange(date) %>%
  mutate(oil_pct_change = c(0, diff(brent) / head(brent, -1) * 100))

data_processed <- data_processed %>%
  { if ("oil_pct_change" %in% names(.)) select(., -oil_pct_change) else . } %>%
  left_join(oil_shock_lookup %>% select(date, oil_pct_change), by = "date")

cat("\nDate range:", as.character(min(data_processed$date)),
    "to", as.character(max(data_processed$date)))

##
## Date range: 2023-01-01 to 2026-06-01

cat("\nCountries:", paste(sort(unique(data_processed$country)), collapse = ", "))

##
## Countries: Canada, India, Indonesia, Poland, South_Africa, Thailand, Turkey, United_S

cat("\nTotal observations:", nrow(data_processed), "\n")

##
## Total observations: 336

```

1.2 Descriptive Statistics

```

data_processed %>%
  select(ln_cpi, ln_ipi, oil_pct_change,
         ln_neer, policy_rate, vix, brent) %>%
  summarise(across(everything(),
    list(Mean = ~round(mean(.), 3), SD = ~round(sd(.), 3),
         Min = ~round(min(.), 3), Max = ~round(max(.), 3)),

```

```

    .names = "{.col}__{.fn}")) %>%
pivot_longer(everything(),
              names_to = c("Variable", "Stat"),
              names_sep = "__") %>%
pivot_wider(names_from = Stat, values_from = value) %>%
mutate(Variable = case_match(Variable,
  "ln_cpi"          ~ "ln(CPI)",
  "ln_ipi"          ~ "ln(IPI)",
  "oil_pct_change" ~ "Oil Shock (%)",
  "ln_neer"         ~ "ln(NEER)",
  "policy_rate"     ~ "Policy Rate (%)",
  "vix"             ~ "VIX",
  "brent"           ~ "Brent (USD/bbl)",
  .default         = Variable
)) %>%
kbl(caption = "Descriptive Statistics - Full Panel (Jan 2023 - Jun 2026)",
    digits = 3, align = "lrrrr", booktabs = TRUE) %>%
kable_styling(latex_options = c("striped", "hold_position"),
              bootstrap_options = c("striped", "hover", "condensed"),
              full_width = FALSE) %>%
footnote(general = "N = 336 observations. 8 countries × 42 months.",
         general_title = "Note:")

```

2 Exploratory Data Analysis

2.1 Brent Crude Oil Price

Table 1: Descriptive Statistics — Full Panel (Jan 2023 – Jun 2026)

Variable	Mean	SD	Min	Max
ln(CPI)	4.866	0.213	4.677	5.685
ln(IPI)	4.721	0.092	4.544	4.909
Oil Shock (%)	0.532	5.964	-11.484	24.968
ln(NEER)	4.266	0.574	2.603	4.835
Policy Rate (%)	8.780	10.795	1.250	50.000
VIX	18.836	5.837	12.440	38.500
Brent (USD/bbl)	80.370	8.732	66.840	101.300

Note:

N = 336 observations. 8 countries $\times$ 42 months.

```
oil_ts <- data_processed %>% distinct(date, brent) %>% arrange(date)

ggplot(oil_ts, aes(x = date, y = brent)) +
  geom_area(fill = "#c0392b", alpha = 0.12) +
  geom_line(color = "#c0392b", linewidth = 1.1) +
  geom_vline(xintercept = as.numeric(as.Date("2026-02-27")),
    linetype = "dashed", color = "#2c3e50", linewidth = 0.9) +
  annotate("label", x = as.numeric(as.Date("2026-02-27")),
    y = max(oil_ts$brent) * 0.88,
    label = "Gulf Crisis\nFeb 27, 2026",
    color = "#2c3e50", size = 3.5, hjust = 1.08,
    fill = "white", label.size = 0.3) +
  scale_x_date(date_breaks = "6 months", date_labels = "%b %Y") +
  scale_y_continuous(labels = dollar_format(prefix = "$")) +
  labs(title = "Brent Crude Oil Price: Pre- and Post-2026 Gulf Crisis",
    subtitle = "Monthly average, USD per barrel",
    x = NULL, y = "USD / Barrel") +
  theme_minimal(base_size = 13) +
  theme(plot.title = element_text(face = "bold"),
```

```
panel.grid.minor = element_blank())
```

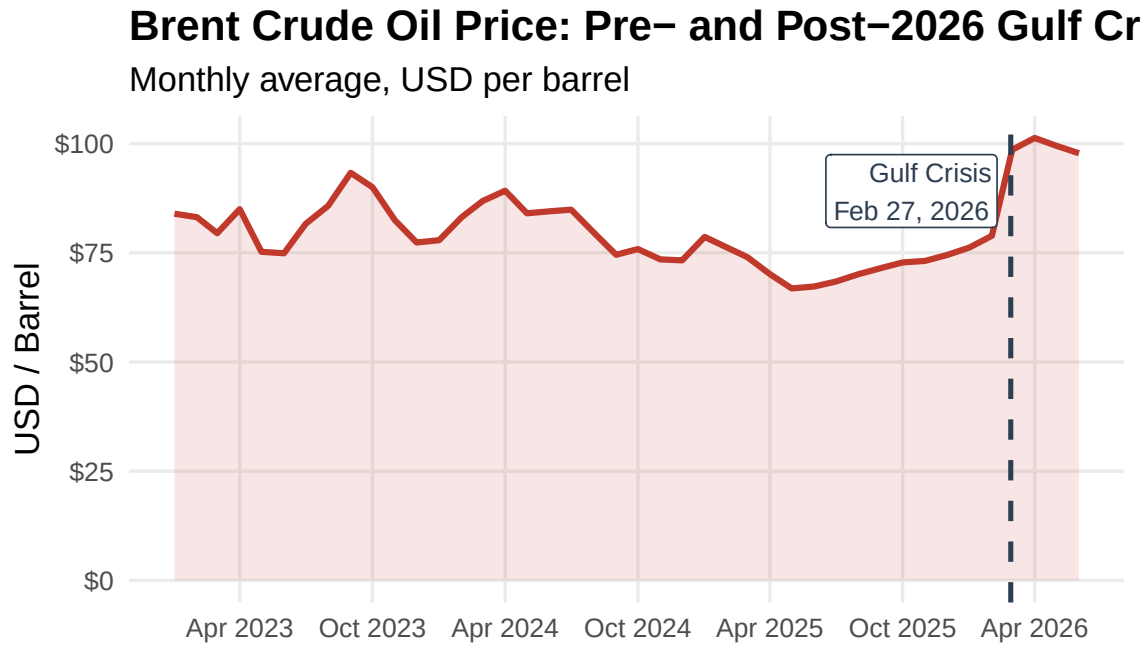


Figure 1: Brent Crude Oil Price: Pre- and Post-2026 Gulf Crisis

2.2 Monthly Oil Shock Variable

```
oil_chg <- oil_shock_lookup %>%
  mutate(direction = ifelse(oil_pct_change >= 0, "Increase", "Decrease"))

ggplot(oil_chg, aes(x = date, y = oil_pct_change, fill = direction)) +
  geom_col() +
  geom_hline(yintercept = 0, linewidth = 0.5) +
  geom_vline(xintercept = as.numeric(as.Date("2026-02-27")),
             linetype = "dashed", color = "#2c3e50") +
  scale_fill_manual(values = c("Increase" = "#27ae60",
                              "Decrease" = "#e74c3c"), name = NULL) +
  scale_x_date(date_breaks = "6 months", date_labels = "%b %Y") +
  labs(title = "Monthly % Change in Brent Crude Oil Price",
       subtitle = "The OilShock variable used in regression; dashed line = Gulf Crisis",
```



```
x = NULL, y = "% Change") +
theme_minimal(base_size = 13) +
theme(plot.title = element_text(face = "bold"),
      legend.position = "none")
```

Monthly % Change in Brent Crude Oil Price

The OilShock variable used in regression; dashed line = Gulf Crisis

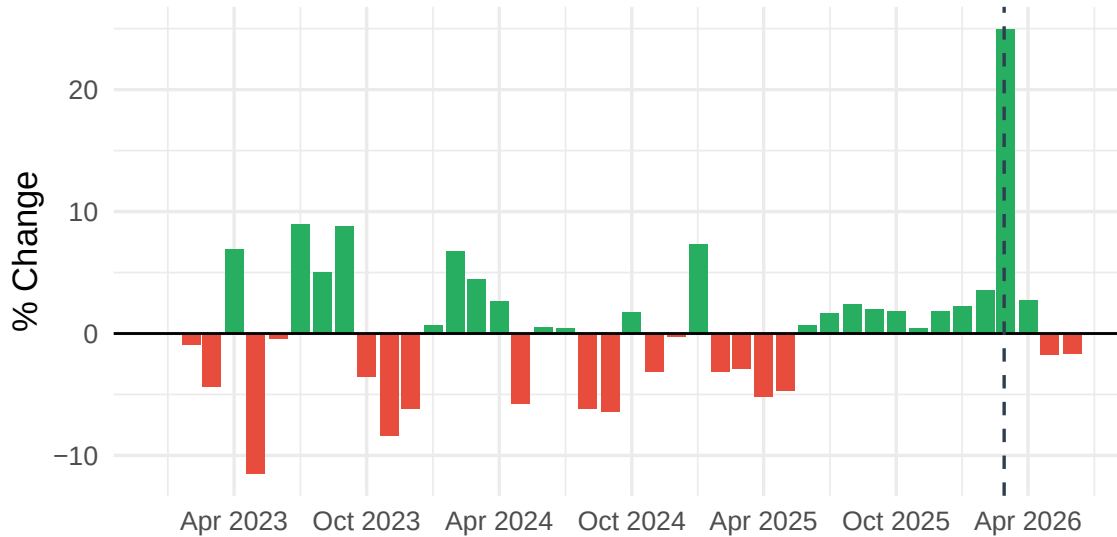


Figure 2: Monthly Percentage Change in Brent Crude (OilShock variable)

2.3 CPI Trends by Country Group

```
avg_cpi <- data_processed %>%
  group_by(date, group) %>%
  summarise(mean_ln_cpi = mean(ln_cpi), .groups = "drop")

ggplot(avg_cpi, aes(x = date, y = mean_ln_cpi, color = group)) +
  geom_line(linewidth = 1.2) +
  geom_vline(xintercept = as.numeric(as.Date("2026-02-27")),
            linetype = "dashed", color = "black") +
  scale_color_manual(values = c("Emerging_Importer" = "#e74c3c",
                                "Advanced_Exporter" = "#2980b9"),
```

```

labels = c("Emerging_Importer" = "Emerging Importers",
           "Advanced_Exporter" = "Advanced Exporters")) +
scale_x_date(date_breaks = "6 months", date_labels = "%b %Y") +
labs(title = "Log CPI Trends by Country Group",
     subtitle = "Group averages; dashed line = Gulf Crisis start (Feb 27, 2026)",
     x = NULL, y = "Log(CPI)", color = NULL) +
theme_minimal(base_size = 13) +
theme(plot.title = element_text(face = "bold"),
      legend.position = "bottom")

```

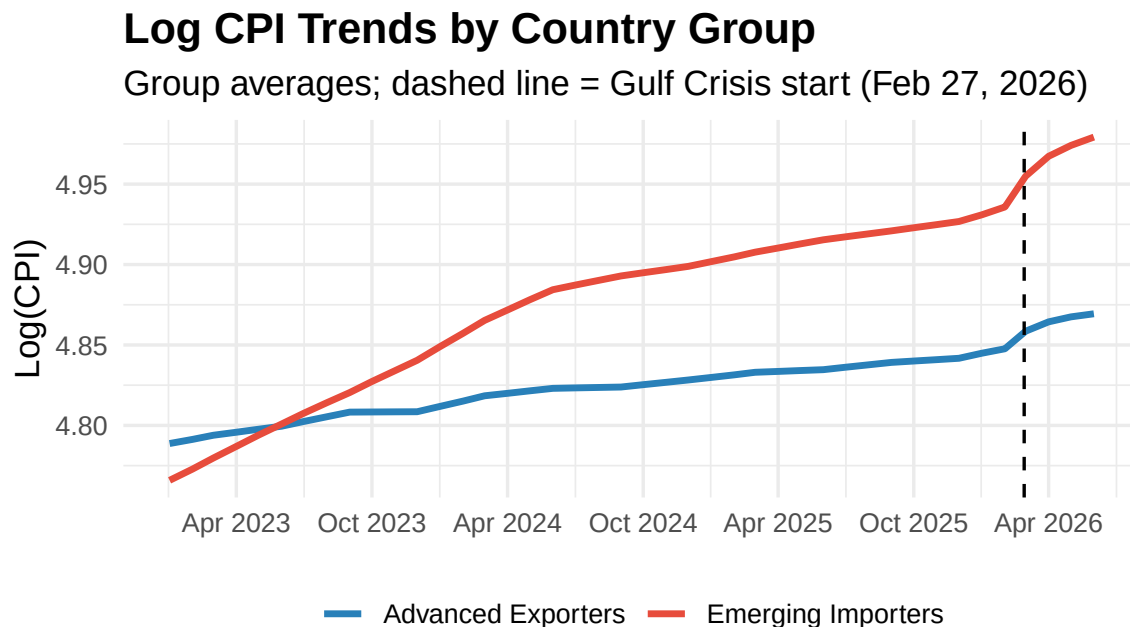


Figure 3: Log CPI Trends: Emerging Importers vs Advanced Exporters

2.4 IPI Trends by Country Group

```

avg_ipi <- data_processed %>%
  group_by(date, group) %>%
  summarise(mean_ln_ipi = mean(ln_ipi), .groups = "drop")

ggplot(avg_ipi, aes(x = date, y = mean_ln_ipi, color = group)) +

```

```
geom_line(linewidth = 1.2) +
geom_vline(xintercept = as.numeric(as.Date("2026-02-27")),
           linetype = "dashed", color = "black") +
scale_color_manual(values = c("Emerging_Importer" = "#e74c3c",
                              "Advanced_Exporter" = "#2980b9"),
                  labels = c("Emerging_Importer" = "Emerging Importers",
                              "Advanced_Exporter" = "Advanced Exporters")) +
scale_x_date(date_breaks = "6 months", date_labels = "%b %Y") +
labs(title = "Log IPI Trends by Country Group",
     subtitle = "Group averages; dashed line = Gulf Crisis start (Feb 27, 2026)",
     x = NULL, y = "Log(IPI)", color = NULL) +
theme_minimal(base_size = 13) +
theme(plot.title = element_text(face = "bold"),
      legend.position = "bottom")
```

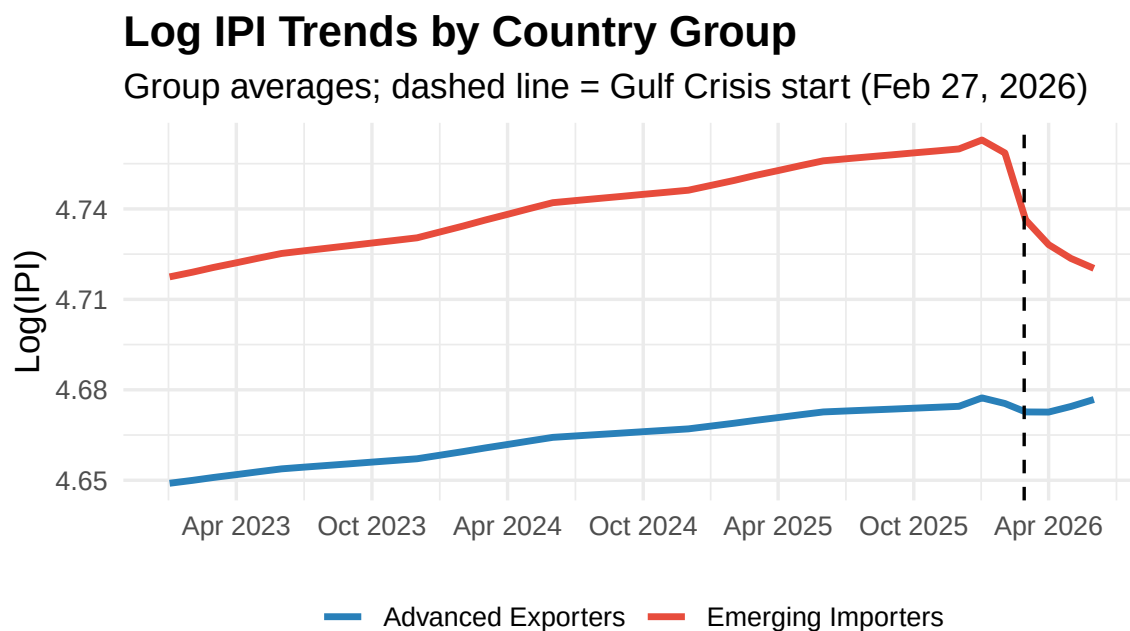


Figure 4: Log IPI Trends: Emerging Importers vs Advanced Exporters

3 ARIMA Forecasting — June 2026

3.1 Stationarity Tests

```
brent_ts <- oil_ts %>% pull(brent) %>% ts(frequency = 12)

cat("=== ADF Test: Brent Oil Price (levels) ===\n")

## === ADF Test: Brent Oil Price (levels) ===

print(adf.test(brent_ts))

##
## Augmented Dickey-Fuller Test
##
## data:  brent_ts
## Dickey-Fuller = -0.94464, Lag order = 3, p-value = 0.9347
## alternative hypothesis: stationary

cat("\n=== ADF Test: Brent Oil Price (1st difference) ===\n")

##
## === ADF Test: Brent Oil Price (1st difference) ===

print(adf.test(diff(brent_ts)))

##
## Augmented Dickey-Fuller Test
##
## data:  diff(brent_ts)
## Dickey-Fuller = -4.4535, Lag order = 3, p-value = 0.01
## alternative hypothesis: stationary
```

3.2 Brent Crude Forecast — June 2026

```
arima_brent <- auto.arima(brent_ts, seasonal = TRUE,
                          stepwise = FALSE, approximation = FALSE,
                          trace = FALSE)
cat("Best ARIMA model for Brent:\n"); print(arima_brent)

## Best ARIMA model for Brent:

## Series: brent_ts
## ARIMA(2,0,0) with non-zero mean
##
## Coefficients:
##          ar1          ar2          mean
##          1.1048   -0.3057   81.4418
## s.e.    0.1444    0.1531    3.3617
##
## sigma^2 = 21.82:  log likelihood = -123.51
## AIC=255.01   AICc=256.09   BIC=261.96

fc_brent <- forecast(arima_brent, h = 1)
autoplot(fc_brent) +
  labs(title = "ARIMA Forecast: Brent Crude Oil - June 2026",
       subtitle = paste0("Model: ARIMA(",
                        paste(arima_brent$arima[c(1,6,2)], collapse = ","), ")"),
       x = "Time (months since Jan 2023)", y = "USD / Barrel") +
  theme_minimal(base_size = 13) +
  theme(plot.title = element_text(face = "bold"))

cat(sprintf("\n-- June 2026 Brent Forecast --\n"))

##
## -- June 2026 Brent Forecast --
```

ARIMA Forecast: Brent Crude Oil — June 2026

Model: ARIMA(2,0,0)

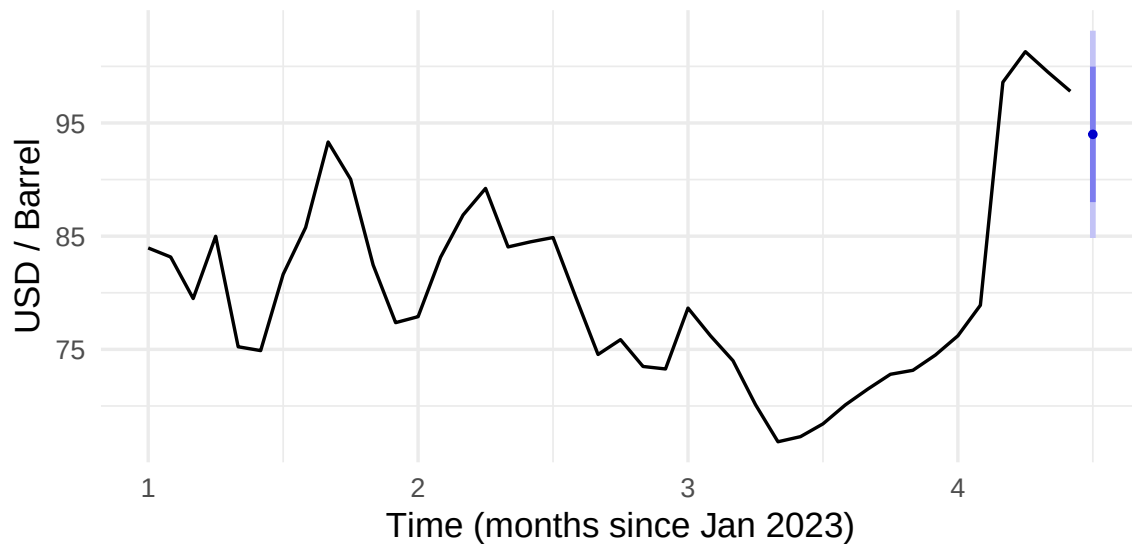


Figure 5: ARIMA Forecast: Brent Crude Oil Price for June 2026

```
cat(sprintf(" Point forecast : $%.2f / barrel\n", fc_brent$mean[1]))
```

```
## Point forecast : $93.99 / barrel
```

```
cat(sprintf(" 80%% PI : [$.2f, $.2f]\n",  
fc_brent$lower[1,1], fc_brent$upper[1,1]))
```

```
## 80% PI : [$88.01, $99.98]
```

```
cat(sprintf(" 95%% PI : [$.2f, $.2f]\n",  
fc_brent$lower[1,2], fc_brent$upper[1,2]))
```

```
## 95% PI : [$84.84, $103.15]
```

3.3 CPI Forecasts — All 8 Countries

```
countries_list <- sort(unique(data_processed$country))
```

```
cpi_forecasts <- tibble(Country = character(), Model = character(),
```

```

        Forecast_CPI = numeric(), PI_80_Low = numeric(),
        PI_80_High = numeric(), PI_95_Low = numeric(),
        PI_95_High = numeric())

for (ctry in countries_list) {
  cpi_ts <- data_processed %>%
    filter(country == ctry) %>% arrange(date) %>%
    pull(ln_cpi) %>%
    ts(frequency = 12, start = c(2023, 1))

  fit <- auto.arima(cpi_ts, seasonal = TRUE,
                    stepwise = FALSE, approximation = FALSE, trace = FALSE)
  fc <- forecast(fit, h = 1)

  cpi_forecasts <- bind_rows(cpi_forecasts, tibble(
    Country      = ctry,
    Model        = paste0("ARIMA(", paste(fit$arima[c(1,6,2)], collapse = ","), ")"),
    Forecast_CPI = round(exp(fc$mean[1]), 4),
    PI_80_Low    = round(exp(fc$lower[1,1]), 4),
    PI_80_High   = round(exp(fc$upper[1,1]), 4),
    PI_95_Low    = round(exp(fc$lower[1,2]), 4),
    PI_95_High   = round(exp(fc$upper[1,2]), 4)
  ))
}

cpi_forecasts %>%
  kbl(caption = "ARIMA Forecasted CPI Level - June 2026",
      col.names = c("Country", "Model", "CPI Forecast",
                    "80% Low", "80% High", "95% Low", "95% High"),
      align = "lcrrrrr", booktabs = TRUE) %>%

```

```
kable_styling(latex_options = c("striped", "hold_position", "scale_down"),
              bootstrap_options = c("striped", "hover", "condensed"),
              full_width = FALSE) %>%
column_spec(3, bold = TRUE, color = "#c0392b")
```

Table 2: ARIMA Forecasted CPI Level — June 2026

Country	Model	CPI Forecast	80% Low	80% High	95% Low	95% High
Canada	ARIMA(1,1,0)	129.0727	128.8019	129.3441	128.6588	129.4880
India	ARIMA(1,1,0)	130.3465	130.0149	130.6789	129.8397	130.8552
Indonesia	ARIMA(1,1,0)	124.9629	124.6638	125.2627	124.5057	125.4217
Poland	ARIMA(1,2,1)	133.1700	132.6945	133.6473	132.4434	133.9007
South_Africa	ARIMA(1,1,0)	129.0664	128.7289	129.4048	128.5505	129.5844
Thailand	ARIMA(0,2,0)	116.2008	115.8784	116.5241	115.7080	116.6956
Turkey	ARIMA(0,2,1)	299.3114	296.6477	301.9991	295.2472	303.4316
United_States	ARIMA(0,1,2)	131.9582	131.6012	132.3161	131.4127	132.5060

3.4 IPI Forecasts — All 8 Countries

```
ipi_forecasts <- tibble(Country = character(), Model = character(),
                        Forecast_IPI = numeric(), PI_80_Low = numeric(),
                        PI_80_High = numeric(), PI_95_Low = numeric(),
                        PI_95_High = numeric())

for (ctry in countries_list) {
  ipi_ts <- data_processed %>%
    filter(country == ctry) %>% arrange(date) %>%
    pull(ln_ipi) %>%
    ts(frequency = 12, start = c(2023, 1))

  fit <- auto.arima(ipi_ts, seasonal = TRUE,
                    stepwise = FALSE, approximation = FALSE, trace = FALSE)
  fc <- forecast(fit, h = 1)
```



```

ipi_forecasts <- bind_rows(ipi_forecasts, tibble(
  Country      = ctry,
  Model        = paste0("ARIMA(", paste(fit$arma[c(1,6,2)], collapse = ","), ")"),
  Forecast_IPI = round(exp(fc$mean[1]), 4),
  PI_80_Low    = round(exp(fc$lower[1,1]), 4),
  PI_80_High   = round(exp(fc$upper[1,1]), 4),
  PI_95_Low    = round(exp(fc$lower[1,2]), 4),
  PI_95_High   = round(exp(fc$upper[1,2]), 4)
))
}

ipi_forecasts %>%
  kbl(caption = "ARIMA Forecasted IPI Level - June 2026",
      col.names = c("Country", "Model", "IPI Forecast",
                    "80% Low", "80% High", "95% Low", "95% High"),
      align = "lcrrrrr", booktabs = TRUE) %>%
  kable_styling(latex_options = c("striped", "hold_position", "scale_down"),
                bootstrap_options = c("striped", "hover", "condensed"),
                full_width = FALSE) %>%
  column_spec(3, bold = TRUE, color = "#27ae60")

```

Table 3: ARIMA Forecasted IPI Level — June 2026

Country	Model	IPI Forecast	80% Low	80% High	95% Low	95% High
Canada	ARIMA(0,1,2)	109.8310	109.6750	109.9872	109.5925	110.0701
India	ARIMA(0,2,2)	131.0283	130.5457	131.5127	130.2909	131.7699
Indonesia	ARIMA(0,2,2)	114.6937	114.2324	115.1568	113.9890	115.4027
Poland	ARIMA(0,2,2)	115.0029	114.4249	115.5838	114.1201	115.8925
South_Africa	ARIMA(2,0,0)	94.1505	93.7282	94.5747	93.5054	94.8001
Thailand	ARIMA(0,2,2)	102.2478	101.7860	102.7118	101.5423	102.9583
Turkey	ARIMA(0,2,2)	118.2893	117.5603	119.0229	117.1762	119.4130
United_States	ARIMA(0,1,1)	105.2029	105.0512	105.3549	104.9710	105.4354

4 Panel Regression — Main Group: Emerging Importers

4.1 Declare Panel Frame

```
pdata_emg <- data_processed %>%  
  filter(country %in% emerging_importers) %>%  
  pdata.frame(index = c("country", "date"))  
  
cat("Panel: Emerging Importers\n")
```

```
## Panel: Emerging Importers
```

```
cat("Countries :", pdim(pdata_emg)$nT$n, "\n")
```

```
## Countries : 6
```

```
cat("Periods :", pdim(pdata_emg)$nT$T, "\n")
```

```
## Periods : 42
```

```
cat("Balanced :", pdim(pdata_emg)$balanced, "\n")
```

```
## Balanced : TRUE
```

```
cat("Observations:", pdim(pdata_emg)$nT$n * pdim(pdata_emg)$nT$T, "\n")
```

```
## Observations: 252
```

4.2 CPI Models — Emerging Importers

```
formula_cpi <- ln_cpi ~ oil_pct_change + ln_neer +  
  policy_rate + vix + post_crisis
```

```

pols_cpi_emg <- plm(formula_cpi, data = pdata_emg, model = "pooling")
fe_cpi_emg   <- plm(formula_cpi, data = pdata_emg, model = "within")
re_cpi_emg   <- plm(formula_cpi, data = pdata_emg, model = "random")

stargazer(pols_cpi_emg, fe_cpi_emg, re_cpi_emg,
  title          = "CPI Regression - Emerging Importers",
  column.labels  = c("Pooled OLS", "Fixed Effects", "Random Effects"),
  dep.var.labels = "ln(CPI)",
  covariate.labels = c("Oil Shock (%  $\Delta$  Brent)",
    "ln(NEER)", "Policy Rate (%)",
    "VIX", "Post-Crisis Dummy"),
  keep.stat      = c("n", "rsq", "adj.rsq"),
  type           = "latex", style = "aer",
  digits         = 6,
  digits.extra   = 4,
  notes          = "Arellano HC3 robust SE applied in Section 8.",
  notes.align    = "l")

```

% Table created by stargazer v.5.2.3 by Marek Hlavac, Social Policy Institute. E-mail: marek.hlavac at gmail.com % Date and time: Sat, May 30, 2026 - 21:28:13

4.3 IPI Models — Emerging Importers

```

formula_ipi <- ln_ipi ~ oil_pct_change + ln_neer +
  policy_rate + vix + post_crisis

pols_ipi_emg <- plm(formula_ipi, data = pdata_emg, model = "pooling")
fe_ipi_emg   <- plm(formula_ipi, data = pdata_emg, model = "within")
re_ipi_emg   <- plm(formula_ipi, data = pdata_emg, model = "random")

```

Table 4: CPI Regression — Emerging Importers

	ln(CPI)		
	Pooled OLS	Fixed Effects	Random Effects
	(1)	(2)	(3)
Oil Shock (	(0.000871)	(0.000205)	(0.000542)
ln(NEER)	−0.149519*** (0.019754)	−1.063687*** (0.017476)	−0.571052*** (0.030935)
Policy Rate (	(0.000988)	(0.000294)	(0.000747)
VIX	0.004182*** (0.001477)	0.001462*** (0.000350)	0.002504*** (0.000923)
Post-Crisis Dummy	0.038096 (0.029355)	0.012104* (0.006928)	0.029420 (0.018294)
Constant	5.291242*** (0.097565)		7.149220*** (0.137195)
Observations	252	252	252
R ²	0.899460	0.965483	0.782483
Adjusted R ²	0.897417	0.964051	0.778062

Notes:

***Significant at the 1 percent level.

**Significant at the 5 percent level.

*Significant at the 10 percent level.

Arellano HC3 robust SE applied in Section 8.

```

stargazer(pols_ipi_emg, fe_ipi_emg, re_ipi_emg,
          title          = "IPI Regression - Emerging Importers",
          column.labels  = c("Pooled OLS", "Fixed Effects", "Random Effects"),
          dep.var.labels = "ln(IPI)",
          covariate.labels = c("Oil Shock (%  $\Delta$  Brent)",
                                "ln(NEER)", "Policy Rate (%)",
                                "VIX", "Post-Crisis Dummy"),
          keep.stat      = c("n", "rsq", "adj.rsq"),
          type           = "latex", style = "aer",
          digits         = 6,
          digits.extra   = 4,
          notes          = "Arellano HC3 robust SE applied in Section 8.",
          notes.align    = "l")

```

% Table created by stargazer v.5.2.3 by Marek Hlavac, Social Policy Institute. E-mail: marek.hlavac at gmail.com % Date and time: Sat, May 30, 2026 - 21:28:15

5 Panel Regression — Comparison Group: Advanced Exporters

5.1 Declare Panel Frame

```

pdata_adv <- data_processed %>%
  filter(country %in% advanced_exporters) %>%
  pdata.frame(index = c("country", "date"))

cat("Panel: Advanced Exporters\n")

```

```
## Panel: Advanced Exporters
```

Table 5: IPI Regression — Emerging Importers

	ln(IPI)		
	Pooled OLS	Fixed Effects	Random Effects
	(1)	(2)	(3)
Oil Shock (	(0.001045)	(0.000149)	(0.000148)
ln(NEER)	0.041229* (0.023715)	−0.072448*** (0.012675)	−0.071686*** (0.012546)
Policy Rate (	(0.001186)	(0.000214)	(0.000213)
VIX	0.001889 (0.001773)	0.001131*** (0.000254)	0.001133*** (0.000253)
Post-Crisis Dummy	−0.038293 (0.035242)	−0.038245*** (0.005024)	−0.038219*** (0.005011)
Constant	4.492407*** (0.117130)		5.016894*** (0.073608)
Observations	252	252	252
R ²	0.106738	0.312957	0.309667
Adjusted R ²	0.088582	0.284449	0.295636

Notes:

***Significant at the 1 percent level.

**Significant at the 5 percent level.

*Significant at the 10 percent level.

Arellano HC3 robust SE applied in Section 8.

```
cat("Countries :", pdim(pdata_adv)$nT$n, "\n")
```

```
## Countries : 2
```

```
cat("Periods :", pdim(pdata_adv)$nT$T, "\n")
```

```
## Periods : 42
```

```
cat("Balanced :", pdim(pdata_adv)$balanced, "\n")
```

```
## Balanced : TRUE
```

```
cat("Observations:", pdim(pdata_adv)$nT$n * pdim(pdata_adv)$nT$T, "\n")
```

```
## Observations: 84
```

5.2 CPI Models — Advanced Exporters

```
pols_cpi_adv <- plm(formula_cpi, data = pdata_adv, model = "pooling")
```

```
fe_cpi_adv <- plm(formula_cpi, data = pdata_adv, model = "within")
```

```
re_cpi_adv <- plm(formula_cpi, data = pdata_adv, model = "random",  
                 random.method = "walhus") # or "amemiya"
```

```
stargazer(pols_cpi_adv, fe_cpi_adv, re_cpi_adv,  
          title          = "CPI Regression - Advanced Exporters",  
          column.labels  = c("Pooled OLS", "Fixed Effects", "Random Effects"),  
          dep.var.labels = "ln(CPI)",  
          covariate.labels = c("Oil Shock (%  $\Delta$  Brent)",  
                               "ln(NEER)", "Policy Rate (%)",  
                               "VIX", "Post-Crisis Dummy"),  
          keep.stat      = c("n", "rsq", "adj.rsq"),  
          type           = "latex", style = "aer",  
          digits         = 6,  
          digits.extra   = 4,
```

```

notes          = "Arellano HC3 robust SE applied in Section 8.",
notes.align    = "l")

```

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Table 6: CPI Regression — Advanced Exporters

	ln(CPI)		
	Pooled OLS	Fixed Effects	Random Effects
	(1)	(2)	(3)
Oil Shock (	(0.000207)	(0.000210)	(0.000207)
ln(NEER)	0.116984*** (0.009617)	0.038297 (0.106160)	0.116984*** (0.009617)
Policy Rate (	(0.001748)	(0.001758)	(0.001748)
VIX	−0.001097*** (0.000406)	−0.001027** (0.000418)	−0.001097*** (0.000406)
Post-Crisis Dummy	0.043040*** (0.007155)	0.043860*** (0.007260)	0.043040*** (0.007155)
Constant	4.367288*** (0.040992)		4.367288*** (0.040992)
Observations	84	84	84
R ²	0.792311	0.743938	0.792311
Adjusted R ²	0.778998	0.723985	0.778998

Notes:

***Significant at the 1 percent level.

**Significant at the 5 percent level.

*Significant at the 10 percent level.

Arellano HC3 robust SE applied in Section 8.

5.3 IPI Models — Advanced Exporters


```

pols_ipi_adv <- plm(formula_ipi, data = pdata_adv, model = "pooling")
fe_ipi_adv  <- plm(formula_ipi, data = pdata_adv, model = "within")
re_ipi_adv  <- plm(formula_ipi, data = pdata_adv, model = "random",
                    random.method = "walhus") # or "amemiya"

stargazer(pols_ipi_adv, fe_ipi_adv, re_ipi_adv,
           title          = "IPI Regression - Advanced Exporters",
           column.labels  = c("Pooled OLS", "Fixed Effects", "Random Effects"),
           dep.var.labels = "ln(IPI)",
           covariate.labels = c("Oil Shock (%  $\Delta$  Brent)",
                                "ln(NEER)", "Policy Rate (%)",
                                "VIX", "Post-Crisis Dummy"),
           keep.stat      = c("n", "rsq", "adj.rsq"),
           type           = "latex", style = "aer",
           digits         = 6,
           digits.extra   = 4,
           notes          = "Arellano HC3 robust SE applied in Section 8.",
           notes.align    = "l")

```

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marek.hlavac at gmail.com % Date and time: Sat, May 30, 2026 - 21:28:18

6 Diagnostic Tests and Robust Standard Errors

6.1 Hausman Test (FE vs. RE)

```

# Emerging Importers
haus_cpi_emg <- phptest(fe_cpi_emg, re_cpi_emg)
haus_ipi_emg <- phptest(fe_ipi_emg, re_ipi_emg)

```

Table 7: IPI Regression — Advanced Exporters

	ln(IPI)		
	Pooled OLS	Fixed Effects	Random Effects
	(1)	(2)	(3)
Oil Shock (	(0.000084)	(0.000084)	(0.000084)
ln(NEER)	−0.107936*** (0.003908)	−0.043909 (0.042669)	−0.107936*** (0.003908)
Policy Rate (	(0.000710)	(0.000707)	(0.000710)
VIX	−0.000415** (0.000165)	−0.000472*** (0.000168)	−0.000415** (0.000165)
Post-Crisis Dummy	0.010834*** (0.002908)	0.010166*** (0.002918)	0.010834*** (0.002908)
Constant	5.215130*** (0.016658)		5.215130*** (0.016658)
Observations	84	84	84
R ²	0.957648	0.763519	0.957648
Adjusted R ²	0.954933	0.745092	0.954933

Notes:

***Significant at the 1 percent level.

**Significant at the 5 percent level.

*Significant at the 10 percent level.

Arellano HC3 robust SE applied in Section 8.

```

cat("=== Hausman Test - CPI (Emerging Importers) ===\n")

## === Hausman Test - CPI (Emerging Importers) ===

print(haus_cpi_emg)

##
## Hausman Test
##
## data: formula_cpi
## chisq = 468.97, df = 5, p-value < 2.2e-16
## alternative hypothesis: one model is inconsistent

cat(if (haus_cpi_emg$p.value < 0.05)
  "\n-> p < 0.05: Reject H0. Fixed Effects preferred.\n"
  else
  "\n-> p >= 0.05: Fail to reject H0. Random Effects acceptable.\n")

##
## -> p < 0.05: Reject H0. Fixed Effects preferred.

cat("\n=== Hausman Test - IPI (Emerging Importers) ===\n")

##
## === Hausman Test - IPI (Emerging Importers) ===

print(haus_ipi_emg)

##
## Hausman Test
##
## data: formula_ipi
## chisq = 0.18177, df = 5, p-value = 0.9993
## alternative hypothesis: one model is inconsistent

```

```
cat(if (haus_ipi_emg$p.value < 0.05)
  "\n-> p < 0.05: Reject H0. Fixed Effects preferred.\n"
  else
  "\n-> p >= 0.05: Fail to reject H0. Random Effects acceptable.\n")
```

```
##
```

```
## -> p >= 0.05: Fail to reject H0. Random Effects acceptable.
```

```
# Advanced Exporters
```

```
haus_cpi_adv <- phptest(fe_cpi_adv, re_cpi_adv)
```

```
haus_ipi_adv <- phptest(fe_ipi_adv, re_ipi_adv)
```

```
cat("\n=== Hausman Test - CPI (Advanced Exporters) ===\n")
```

```
##
```

```
## === Hausman Test - CPI (Advanced Exporters) ===
```

```
print(haus_cpi_adv)
```

```
##
```

```
## Hausman Test
```

```
##
```

```
## data: formula_cpi
```

```
## chisq = 0.55394, df = 5, p-value = 0.99
```

```
## alternative hypothesis: one model is inconsistent
```

```
cat(if (haus_cpi_adv$p.value < 0.05)
  "\n-> p < 0.05: Reject H0. Fixed Effects preferred.\n"
  else
  "\n-> p >= 0.05: Fail to reject H0. Random Effects acceptable.\n")
```

```
##
```

```
## -> p >= 0.05: Fail to reject H0. Random Effects acceptable.
```

```

cat("\n=== Hausman Test - IPI (Advanced Exporters) ===\n")

##

## === Hausman Test - IPI (Advanced Exporters) ===

print(haus_ipi_adv)

##

## Hausman Test

##

## data: formula_ipi
## chisq = 2.2706, df = 5, p-value = 0.8106
## alternative hypothesis: one model is inconsistent

cat(if (haus_ipi_adv$p.value < 0.05)
  "\n-> p < 0.05: Reject H0. Fixed Effects preferred.\n"
  else
  "\n-> p >= 0.05: Fail to reject H0. Random Effects acceptable.\n")

##

## -> p >= 0.05: Fail to reject H0. Random Effects acceptable.

```

6.2 Breusch-Pagan LM Test (RE vs. POLS)

```

bp_cpi_emg <- plmtest(pols_cpi_emg, effect = "individual", type = "bp")
bp_ipi_emg <- plmtest(pols_ipi_emg, effect = "individual", type = "bp")
bp_cpi_adv <- plmtest(pols_cpi_adv, effect = "individual", type = "bp")
bp_ipi_adv <- plmtest(pols_ipi_adv, effect = "individual", type = "bp")

cat("=== BP Test - CPI (Emerging Importers) ===\n"); print(bp_cpi_emg)

## === BP Test - CPI (Emerging Importers) ===

##

```

```

## Lagrange Multiplier Test - (Breusch-Pagan)
##
## data: formula_cpi
## chisq = 458.69, df = 1, p-value < 2.2e-16
## alternative hypothesis: significant effects

cat("\n=== BP Test - IPI (Emerging Importers) ===\n"); print(bp_ipi_emg)

##
## === BP Test - IPI (Emerging Importers) ===
##
## Lagrange Multiplier Test - (Breusch-Pagan)
##
## data: formula_ipi
## chisq = 4626.1, df = 1, p-value < 2.2e-16
## alternative hypothesis: significant effects

cat("\n=== BP Test - CPI (Advanced Exporters) ===\n"); print(bp_cpi_adv)

##
## === BP Test - CPI (Advanced Exporters) ===
##
## Lagrange Multiplier Test - (Breusch-Pagan)
##
## data: formula_cpi
## chisq = 1.0205, df = 1, p-value = 0.3124
## alternative hypothesis: significant effects

cat("\n=== BP Test - IPI (Advanced Exporters) ===\n"); print(bp_ipi_adv)

##
## === BP Test - IPI (Advanced Exporters) ===

```

```
##
##  Lagrange Multiplier Test - (Breusch-Pagan)
##
## data:  formula_ipi
## chisq = 1.0088, df = 1, p-value = 0.3152
## alternative hypothesis: significant effects
```

6.3 Robust Standard Errors

```
cat("=== FE CPI - Emerging Importers | Arellano HC3 ===\n")

## === FE CPI - Emerging Importers | Arellano HC3 ===

coeftest(fe_cpi_emg, vcov = vcovHC(fe_cpi_emg, method = "arellano", type = "HC3"))

##
## t test of coefficients:
##
##              Estimate Std. Error  t value  Pr(>|t|)
## oil_pct_change -0.00015398  0.00014007  -1.0993  0.2727210
## ln_neer        -1.06368707  0.01925530 -55.2413 < 2.2e-16 ***
## policy_rate     0.00084126  0.00018779   4.4799 1.154e-05 ***
## vix             0.00146211  0.00038789   3.7694 0.0002059 ***
## post_crisis     0.01210387  0.00802271   1.5087 0.1326851
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

cat("\n=== FE IPI - Emerging Importers | Arellano HC3 ===\n")

##
## === FE IPI - Emerging Importers | Arellano HC3 ===
```

```

coeftest(fe_ipi_emg, vcov = vcovHC(fe_ipi_emg, method = "arellano", type = "HC3"))

##
## t test of coefficients:
##
##              Estimate   Std. Error t value  Pr(>|t|)
## oil_pct_change  2.4116e-04  1.8431e-05  13.0847 < 2.2e-16 ***
## ln_neer         -7.2448e-02  1.8652e-02  -3.8842 0.0001327 ***
## policy_rate     1.5526e-04  2.6189e-04   0.5929 0.5538360
## vix             1.1308e-03  1.4687e-04   7.6993 3.525e-13 ***
## post_crisis     -3.8245e-02  4.7702e-03  -8.0174 4.673e-14 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

cat("\n=== FE CPI - Advanced Exporters | Arellano HC3 ===\n")

##
## === FE CPI - Advanced Exporters | Arellano HC3 ===

coeftest(fe_cpi_adv, vcov = vcovHC(fe_cpi_adv, method = "arellano", type = "HC3"))

##
## t test of coefficients:
##
##              Estimate   Std. Error t value  Pr(>|t|)
## oil_pct_change -1.2038e-04  3.3478e-05  -3.5958 0.0005687 ***
## ln_neer         3.8297e-02  4.0244e-02   0.9516 0.3442643
## policy_rate     -1.7030e-02  3.6214e-03  -4.7027 1.107e-05 ***
## vix             -1.0274e-03  5.4704e-05 -18.7806 < 2.2e-16 ***
## post_crisis     4.3860e-02  1.6640e-03  26.3587 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```



```
cat("\n=== FE IPI - Advanced Exporters | Arellano HC3 ===\n")

##
## === FE IPI - Advanced Exporters | Arellano HC3 ===

coeftest(fe_ipi_adv, vcov = vcovHC(fe_ipi_adv, method = "arellano", type = "HC3"))

##
## t test of coefficients:
##
##
##           Estimate   Std. Error t value   Pr(>|t|)
## oil_pct_change -5.3152e-05  1.6132e-05 -3.2948   0.001490 **
## ln_neer        -4.3909e-02  1.9813e-02 -2.2162   0.029633 *
## policy_rate    -9.6894e-03  1.3534e-03 -7.1592  4.147e-10 ***
## vix            -4.7209e-04  7.3418e-05 -6.4302  9.753e-09 ***
## post_crisis     1.0166e-02  3.4023e-03  2.9881   0.003765 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

6.4 Coefficient Comparison Plot

```
extract_coef <- function(model, label) {
  ct <- coeftest(model, vcov = vcovHC(model, method = "arellano", type = "HC3"))
  est <- ct["oil_pct_change", "Estimate"]
  se <- ct["oil_pct_change", "Std. Error"]
  data.frame(Group = label,
             Estimate = est,
             CI_low = est - 1.96 * se,
             CI_high = est + 1.96 * se)
}

coef_df <- rbind(
```

```

extract_coef(fe_cpi_emg, "Emerging Importers\n(Main Group)"),
extract_coef(fe_cpi_adv, "Advanced Exporters\n(Comparison Group)")
)

ggplot(coef_df, aes(x = Group, y = Estimate, color = Group)) +
  geom_hline(yintercept = 0, linetype = "dashed", color = "grey50") +
  geom_errorbar(aes(ymin = CI_low, ymax = CI_high),
               width = 0.15, linewidth = 0.9) +
  geom_point(size = 5) +
  scale_color_manual(
    values = c("Emerging Importers\n(Main Group)" = "#e74c3c",
              "Advanced Exporters\n(Comparison Group)" = "#2980b9")) +
  labs(title = "Oil Shock Coefficient on ln(CPI) by Country Group",
       subtitle = "Fixed Effects model; bars = 95% CI (Arellano HC3)",
       x = NULL, y = "Coefficient Estimate") +
  theme_minimal(base_size = 13) +
  theme(plot.title = element_text(face = "bold"),
        legend.position = "none")

```

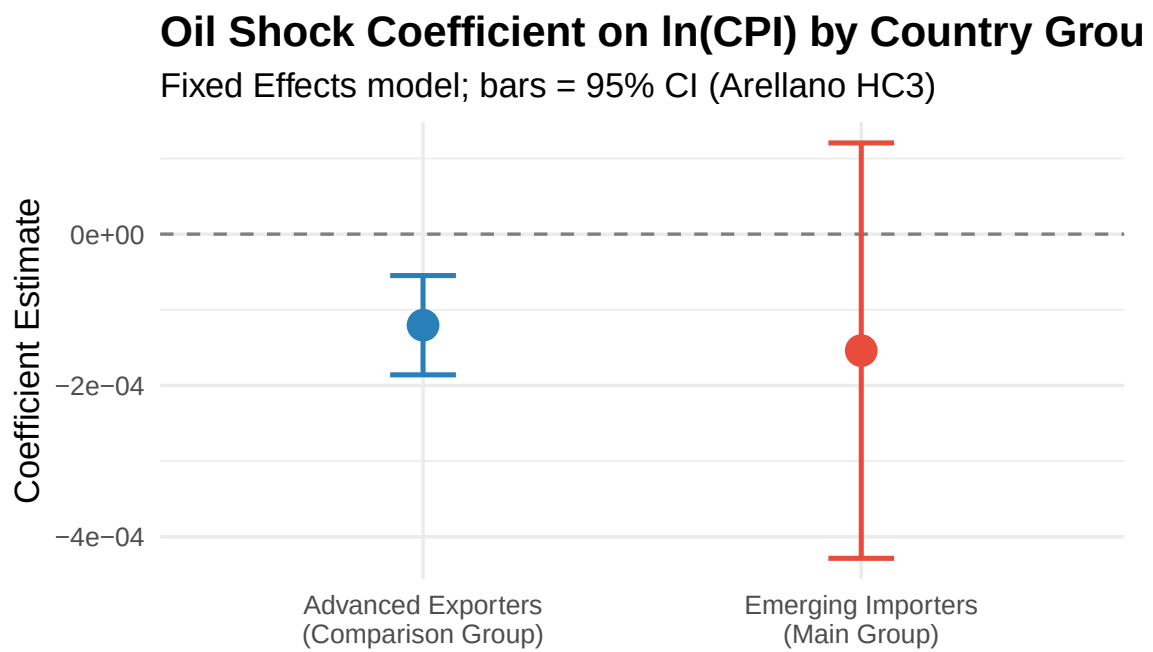


Figure 6: Oil Shock Coefficient on $\ln(\text{CPI})$: Emerging Importers vs Advanced Exporters